Board to Be Wild
Engineering Design Report

Luke Garratt

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1 Introduction

In this project a wooden charcuterie board was designed and produced utilizing several different tools such as Adobe Illustrator, Fusion 360, Aspire, and several hands on machines. The designed charcuterie board depicts two koi fish encircling each other to create a Japanese style.

2 Initial Design and Planning

To plan out the design that will be milled into the wood, a file was created in Adobe Illustrator and an svg of the two koi fish was imported. In Adobe Illusrator the imported design was scaled to match the estimated dimensions of the wooden board. The estimated dimensions were 10 inches by 12 inches by 1 inch. Once the design was finalized in Adobe Illustrator, it was exported as an SVG file and brought into Fusion 360 as a new project. Prior to importing the koi fish design into Fusion 360, a model of the wooden board needed to be created by using the two-point rectangle tool, the extrude feature, and the appearance tool to match the wood's colors.

3 Material Preparation

The board was first cut using the table saw, the board had to be slid quickly through the saw to avoid burn marks, the burn marks would appear because of how fast the blade is spinning and the friction between it and the wood. The first few attempts left some burn marks, but after finding the right speed to move the board, no more burns were created. The next step in the process was to glue the now cut wood pieces together to create the board using wood glue. A disposable paint brush was used to apply the wood glue to both sides of the wood strips before compressing them together. After all of the wood strips were glued together, two clamps were placed on the wood to hold the shape in place for twenty four hours as it dries. The next step in the process was to use the planer. The planer machine allows the front and back faces of the board to be cut to uniform height, and this step is crucial after the wood glue, so the planer cuts away any residue left from the glue.



Figure 1: Initial Wood

4 Digital Design

A new project was created in Fusion360, a two point rectangle was created with the estimated dimensions of the wood strips, in total five were created and modeled to math the estimated dimensions and shape of the wood board. The appearance tool was used to change the material of the wood strips to reflect the colors of the wood being used. The koi fish svg created in Adobe Illustrator was imported and placed on top of the model. The svg was then selected and extruded into the model to a depth of 0.25 in. Once the model was completed and the final product could be visualized, the koi fish svg was imported into the Aspire software. Aspire allows for the creation of tool paths for the CNC machine to follow. In total, two tool paths were created. The first tool path was a 2D pocket that shaped the outline of the koi fish, a relatively small bit was used for maximum detail. The second tool path focused on the interior details of the fish, it used a pocket tool path, and switched to a V-carve bit to allow for different depths and sharp details. The dimensions of the board had to be measured and entered into Aspire. The jig being used had an offset of 5.175 in by 6.175 in, and the safe z height was set to 2 in.

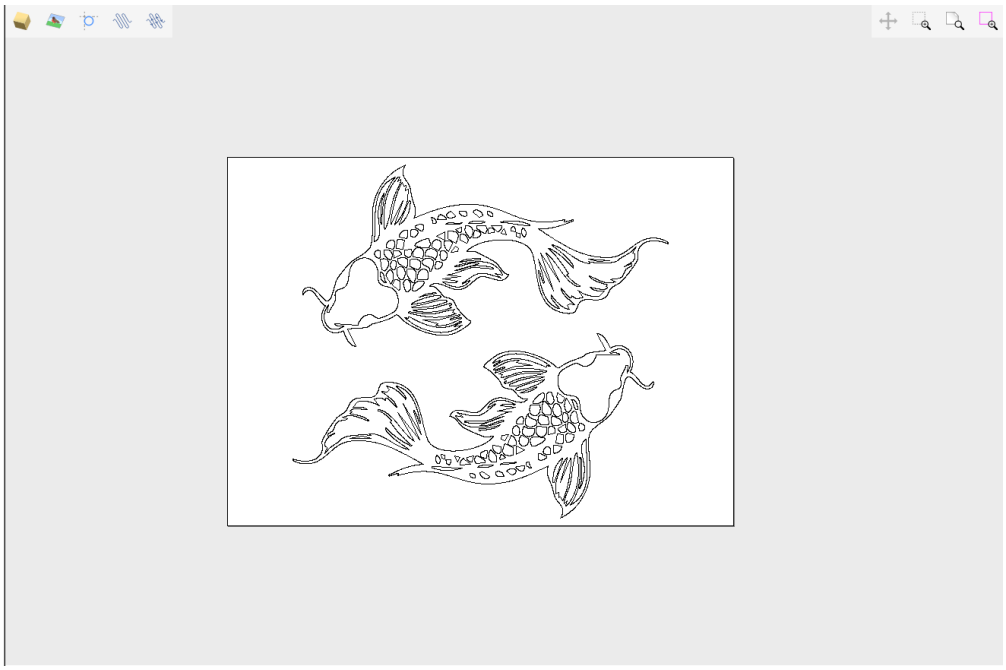


Figure 2: Design in Aspire

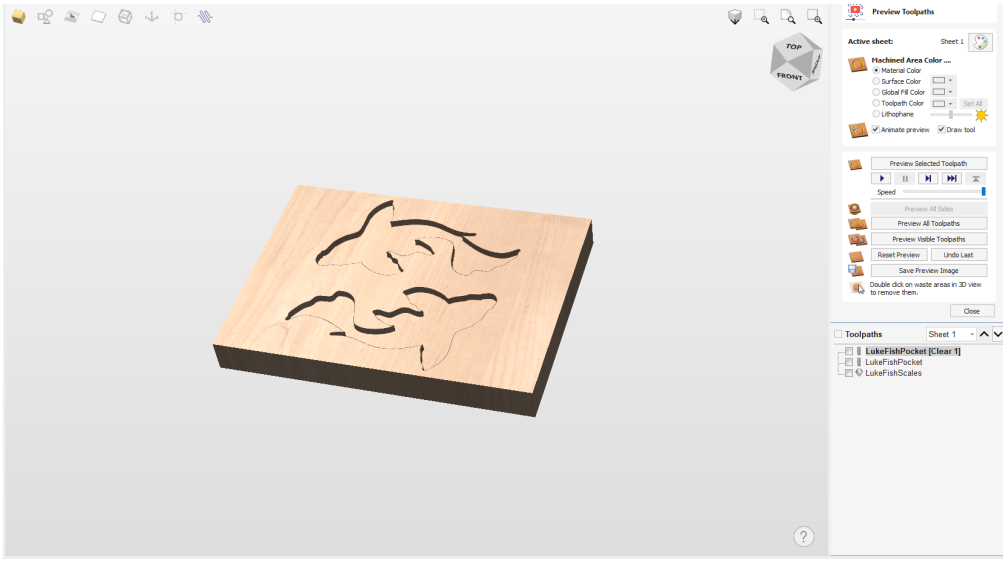


Figure 3: 1st Toolpath

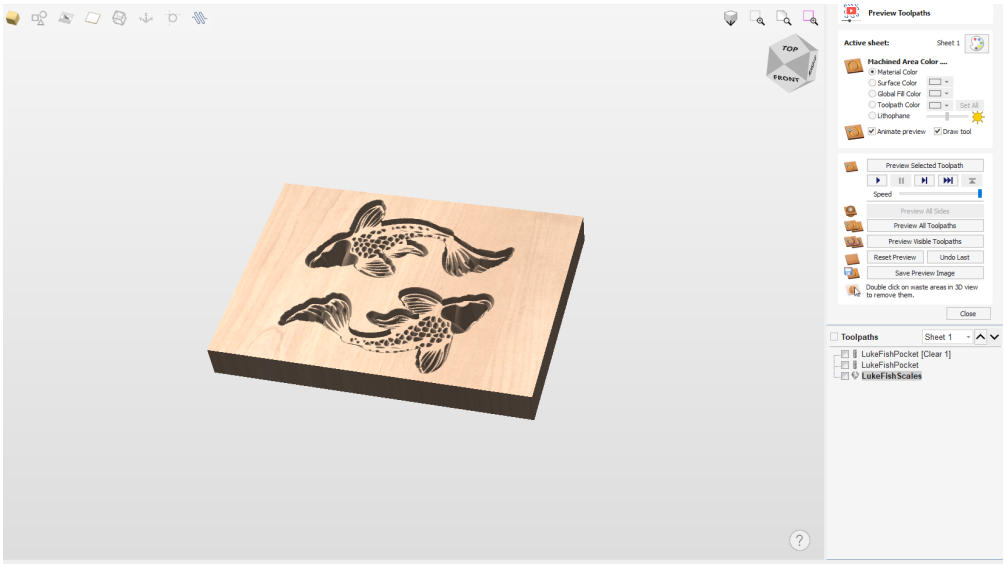


Figure 4: Second Toolpath

5 CNC and Manufacturing

Once the Aspire file was finalized and uploaded to Google Drive, the file was downloaded to a separate computer that was connected to the CNC machine. The Aspire file was rechecked on the new computer to ensure that there would be no errors in the process. The offset did not need to be entered on the machine manually because that step was taken in Aspire before uploading the file. The auto vacuum was connected to the CNC machine to vacuum all the wood shavings that the machine would create. The wood was placed in the jig, and the machine was turned on and the tool paths were run. The entire process took 38 minutes from start to finish. The process that took the largest amount of time was the safe z height. The safe z height could have been set lower to save a significant amount of time.

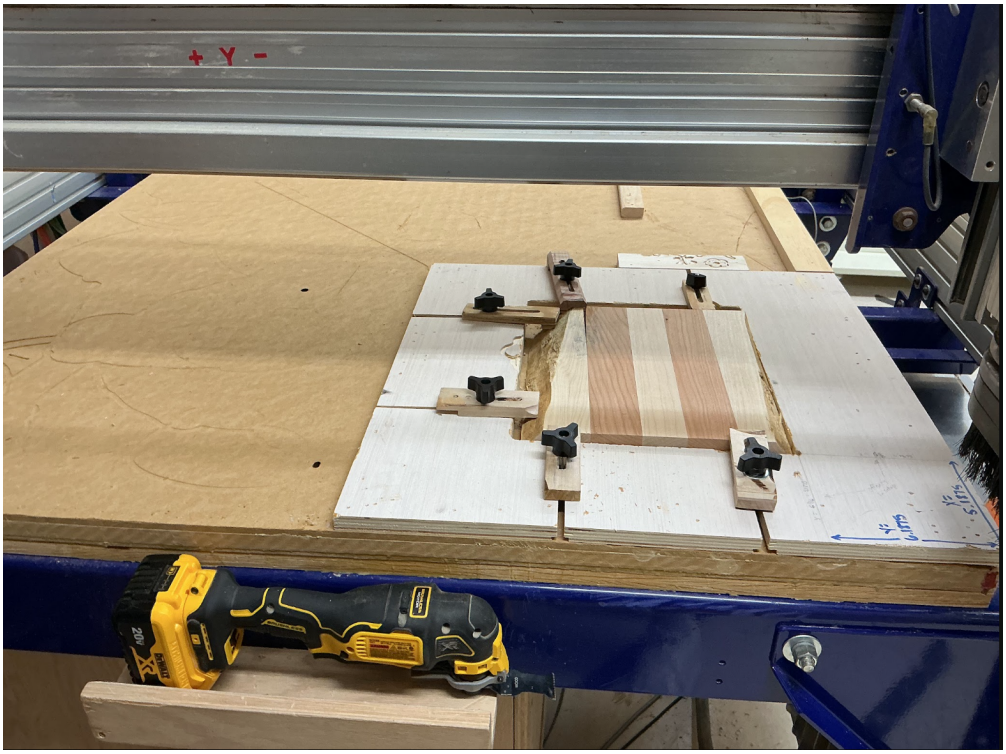


Figure 5: Board in CNC Machine

6 Resin Process

The resin usage was the largest portion of this project. The first layer of resin was the scales and head. In total four colors were used in this step of the resin process. The resin had to first be mixed as clear resin for three minutes, then the necessary colors were added. The resin was applied using a pipette, and colors were mixed together for a more interesting result. The resin was left to dry for 24 hours before adding the second layer of resin. Before adding the second layer of resin a cardboard form had to be created because of the board's live edges to be able to hold the resin. The next layer consisted of clear resin in the fish, and two tones of blue resin on the live edges to fill out the board. The board was painted first with clear resin, and then the colored resin was poured. The second layer was left for another 24 hours. Next, the planer was used to level the board, and the table saw to create square edges. The board was then sanded using an orbital sander and sandpaper increasing in grits, starting at 60 and moving all the way up to 400 grit. The board was then oiled and left to dry for another 24 hours which finished the production process.



Figure 6: First Layer of Resin: (Scales)



Figure 7: Final Layer of Resin

7 Final Product



Figure 8: Final cutting board

8 Reflection

This project was a watershed in the engineering domain. A piece of wood was taken all the way from a plain plank to a stunningly polished result. From the initial idea of incorporate two koi fish circling each other as the focal point of the board to making the idea become reality over the course of three weeks, engineering knowledge was applied through every step of the process. The only challenge faced was the drill bit being too large to give very fine details, but the v-carve bit allowed for close to perfect details on the interior of the fish. Something that could be improved would be using the router on the edges of the board to make them softer, however there was no time to complete this step of the process. Knowledge was gained on how to mix and utilize resin, as well as knowledge on a vast majority of wood working tools and machines which allowed for the completion of major skills in this course. Overall, this project was the most interesting and carefully thought out project in this course, and the skills and knowledge gained from it was invaluable.